Robed Beauvile-Owen, Ph.D.

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EXPERIENCE

Founder & Data Scientist — *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy for retail and service businesses, identifying a $1.5M expansion opportunity.
- Built a revenue driver tree (traffic × capture rate × ticket price) and a margin model that turned client profit goals into break-even targets.
- Estimated the incremental revenue of new sites with observational causal inference (Double ML, causal forests) on a 12-year county panel, with robustness checks (difference-in-differences) and out-of-sample back-tests against real sites.
- Built scenario planning (low/base/high) and Monte Carlo risk models (probability of loss, VaR, CVaR) on reproducible, tested Python workflows (150+ unit tests) that cut turnaround by 91%.

Senior Data Scientist — *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Owned the consumer funnel KPI framework and driver tree linking gross lead rate (GLR) to member rate and cost per member with strategy, finance, and engineering, using a causal graph (DAG) of the multi-metric relationships to show GLR was the lever to model and test.
- Redefined GLR for a new client missing key lead markers, working with data engineering to date each lead by last response, credit it to its campaign, and flag broker influence, establishing a certified metric definition.
- Built uplift and segmentation models on millions of HIPAA-protected records, used in 87% of campaigns, cutting ad spend by $10K per campaign and moving 12% of one client's budget to its top GLR segment.
- Built an in-house experimentation platform that data science and strategy used to validate lift and set campaign budgets, with CUPED cutting metric variance by 20%.
- Found that claims and ticket complaints drove churn, and re-architected the churn prediction platform on Spark and Databricks, cutting runtime by 48% and memory by 23%.
- Owned monthly monitoring of up to 10 production models (accuracy, drift, forecast vs. actual) for leadership and a company-wide self-service analytics dashboard on curated gold tables in Snowflake and Databricks.

Founding Machine Learning Engineer — *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- As the first ML hire, defined metrics and drift monitoring for search and pricing models, and shipped a regression pricing API (C#, Azure, 10,000+ daily requests) that held a stable 18% contribution margin.
- Cut search dissatisfaction by 12% and raised platform engagement by 16% with custom Elasticsearch indexing and TF-IDF weighting.

Data Scientist — *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Defined the core campaign metrics and translated financial goals into team targets with a funnel sensitivity model showing a $1.00 indexed cost per member required GLR of at least 14.2% (each +1 point lowers the index by about 0.07).
- Optimized GLR to drive a 24% increase in paid member conversions within 60 days, and validated GLR out of sample as a leading indicator of member rate 60 days ahead ($R^2 = 0.28$).
- Allocated campaign spend across audiences by eligibility and historical response rates, and built SHAP-explained XGBoost response models that lifted response rates by 3%.

Graduate Researcher & Teaching Assistant — *University of Pittsburgh* Pittsburgh, PA | August 2017 - May 2022

- Produced 3 original results in optimization and operator theory, and taught statistics to classes of 75-200 students.

TECHNICAL PROJECTS

Customer LTV & Churn Survival Modeling github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Showed the LTV = ARPU / churn identity holds only under constant churn, then simulated 1M customer lifetimes with tenure-dependent churn (10%, 5%, then 2% monthly) where the shortcut breaks.
- Estimated expected lifetime and LTV from the empirical survival curve ($E[T] = \int S(t)dt$), matching the simulated mean.

Mobile Game Retention A/B Test github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Analyzed a test moving a mobile game's first progress gate from level 30 to 40, measuring 1- and 7-day retention and rounds played.
- Set Type I/II error rates, sized samples for binary and continuous metrics, removed outliers with IQR, and weighed statistical against practical significance.

Medicare Claims Intelligence & Fraud Detection github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Built a dbt warehouse (4 staging models, 5 marts, 26 data tests) over 790K CMS claims, with tests that caught 11,253 claims missing a service date.
- Beat a seasonal-naive forecast in a 35-origin rolling backtest (MASE 0.62 vs. 1.06), reached 81% precision on planted fraud at review capacity, and built an out-of-time XGBoost risk model (SHAP, MLflow) finding 46% more high-cost members.

Starbucks Rewards: Predicting Offer Responses github.com/beauvilerobed/predict-consumer-responses

- Combined transaction, offer, and demographic data to predict how rewards app members respond to BOGO, discount, and informational offers, with an XGBoost multi-class model trained and deployed on AWS SageMaker and scored on F1 against a benchmark.

EDUCATION

Ph.D. in Mathematics — *University of Pittsburgh* 2017 - Dec 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics — *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90

TECHNICAL SKILLS

Metrics & Planning: KPI Frameworks, KPI / Driver Trees, Certified Metric Definitions, Leading Indicators, Consumer Product Analytics (Acquisition, Activation, Engagement, Retention, Monetization), CAC, LTV / ARPU, Contribution Margin, Sensitivity & Scenario Analysis, Target Setting, Budget Allocation, Forecasting & Backtesting

Causal Inference & Experimentation: A/B Testing, Experimentation Platforms, CUPED, Power Analysis, Observational Causal Methods, Double ML, Causal Forests, Difference-in-Differences, Uplift Modeling, Causal Graphs (DAGs), Robustness Checks

Modeling & Statistics: Regression, Classification, XGBoost, Time Series Forecasting, Survival Analysis, Monte Carlo Simulation, Graph Models, Multi-Metric Modeling, SHAP, Model Monitoring & Drift

Data & Tools: SQL, Python (pandas, statsmodels, scikit-learn, econml), Snowflake, Databricks, Spark, dbt, Gold Tables & Data Marts, Self-Service Analytics, Reproducible & Tested Workflows, AWS, Azure, Power BI, Tableau, Git

Collaboration: Cross-Functional Partnership (Product, Finance, Engineering, Strategy), Stakeholder Communication, Method Documentation, Mentoring